



Special Focus

The Effects of Geopolitical Oil Supply Shocks

The last few years have seen more frequent surges in geopolitical tensions, as exemplified by the recent outbreak of war in the Middle East. Many of these surges have led to sharp increases in oil prices which, if sustained, can create severe headwinds for the global economy. This special focus examines the behavior of oil markets during periods of rising geopolitical risk and quantifies the effects of geopolitical oil supply shocks, as distinct from other sources of oil supply variation, drawing on a novel methodology. It estimates that during times of surging geopolitical risk, a 1 percent reduction in oil production generates a peak oil price increase, on average, of more than 11 percent, nearly twice the increase reported in previous studies for oil supply shocks in general. Outsized oil price responses to geopolitical supply shocks suggest additional layers of precautionary behavior in oil markets, which can have different forms over time. First, at the onset of a significant geopolitical oil shock, a surge in uncertainty over future supplies and global economic conditions can drive risk premia higher, reflecting both efforts to secure physical stocks and speculative market responses. Second, the experience of geopolitical disruptions prompts greater inventory building in the medium-term as a form of self-insurance. Oil supply shocks arising out of conflict have significant and persistent spillover effects on other commodity markets—particularly natural gas and fertilizers. Overall, geopolitically driven oil supply disruptions tend to be especially destabilizing for commodity markets, with likely knock-on consequences for inflation, growth, and poverty. These results highlight the importance of policy measures to mitigate exposure to such shocks.

Introduction

Geopolitical developments often play a central role in shaping global oil market movements. The recent outbreak of conflict in the Middle East and associated oil price surge have once again underscored this relationship. However, in recent decades, many episodes have highlighted the strength and speed of the link from surges in geopolitical risk to steep rises in oil prices. Most commonly, heightened tensions in oil-producing regions have disrupted actual or expected supply, putting upward pressure on prices, but geopolitical developments that weaken global demand can instead push prices downward. Importantly, oil prices, like flexible prices in other commodity and asset markets, reflect expectations about future market conditions, and may rise in the absence of actual supply disruptions if market participants reduce their expectations about future supply. Whether such price increases endure will depend in large part on whether the anticipated shift materializes.

Broadly, three types of developments affecting supply can drive oil price movements. First, there can be episodes of heightened geopolitical tensions that lead to actual, previously unexpected, physical

disruptions in oil supply. These tend to have the largest effects on prices. Iraq's invasion of Kuwait in August 1990 provides an example—global oil production fell by about 6 percent, and oil prices surged by roughly 33 percent within a month. The conflict in the Gulf region of the Middle East, which started in March 2026, is another such instance and may amount to the largest supply shock on record, with estimates by the International Energy Agency (IEA) suggesting a reduction of close to 9 percent in global oil supply by early April 2026 (IEA 2026). Correspondingly, the average Brent oil price climbed by 46 percent in March 2026, relative to February.

Second, there can be episodes of elevated geopolitical tensions that do not involve actual supply disruptions but nonetheless trigger price responses in anticipation of possible supply disruptions. For instance, when armed conflict unexpectedly erupted in the Middle East in October 2023, oil prices quickly increased by about 4 percent despite no immediate changes in oil production.

Third, oil prices can respond to changes in supply conditions signaled by developments other than geopolitical tremors. These include revisions in estimates of global reserves—resulting, for example, from new oil discoveries—and announcements by OPEC of changes in

Note: This special focus was prepared by Guillermo Verduzco-Bustos and Phil Kenworthy.

production agreements. The anticipation of such developments may mean that they are reflected in prices before they occur. Thus, in November 2017, OPEC's announcement of the extension of production cuts of around 2 percent led to an increase in oil prices of only 0.2 percent, since the decision had already been largely priced in by markets.

Experience with the first two types of episodes shows how geopolitical developments threatening oil supplies can lead to sharp increases in the level and volatility of oil prices—whether through actual or expected supply disruptions or heightened uncertainty about future supplies. This special focus analyzes oil market dynamics over the past 40 years by identifying geopolitical oil supply shocks, defined as oil supply movements that arise around large increases in geopolitical risk, and analyzing their effects on oil prices and spillovers to other commodity markets. The analysis addresses three questions:

- What are the characteristics of oil price movements around periods of rising geopolitical tensions?
- What are the oil market effects of oil supply shocks associated with surges in geopolitical risk?
- How do geopolitical oil supply shocks propagate to other commodity markets?

Main findings

The special focus presents the following main findings:

Elevated oil market volatility coincides with geopolitical tensions. Oil price volatility, measured by the standard deviation of monthly price changes, has on average been roughly twice as high during periods of rising perceptions of geopolitical risk as during normal market conditions. Episodes of rising geopolitical risk are typically characterized by market assessments of increased risks of supply disruptions, together with greater uncertainty about current and future oil market conditions. Price volatility is naturally higher during such periods.

Geopolitical oil supply shocks lead to larger price effects than other types of oil supply shocks. A 1 percent decline in oil production associated with a geopolitical shock is estimated to result, on average, in an 11.5 percent increase in oil prices. This is roughly twice as large as the typical price response to oil supply shocks estimated in earlier studies (Caldara et al. 2019; Baumeister and Hamilton 2019). This finding reflects the novel identification strategy employed here. By isolating oil price surprises that occur during periods of heightened geopolitical risk, the analysis captures the particular oil market dynamics associated with heightened uncertainty and geopolitical threats, and finds that oil price responses to supply shocks are larger in these settings than at other times.

Geopolitical oil supply shocks have had distinct effects on the behavior of oil inventories. As with other oil supply shocks, oil inventories have initially been drawn down to offset reduced production. However, with geopolitical, but not other, supply shocks, this initial decline has typically been subsequently reversed, with inventories increasing beyond their original level and remaining persistently elevated. This is likely to reflect precautionary stock-building by market participants whose assessment of future supply risks and broader economic uncertainty has been increased by their experience of geopolitical shocks.

*Geopolitical oil supply shocks have had significant spillovers to other commodity markets, with marked increases in commodity prices, particularly for natural gas and fertilizers.*¹ A 10 percent oil price increase due to a geopolitical oil supply shock has been associated, on average, with increases in natural gas prices that have peaked at about 7 percent after 11 months, and increases in fertilizer prices that have peaked at a little over 5 percent after 12 months. These price increases have tended to show notable persistence. Prices in other commodity groups—including food, agricultural raw materials, and precious metals—have also responded significantly to these shocks, but with smaller increases. The estimates also suggest that following geopolitical oil supply shocks, the prices of

¹ For a detailed discussion about commodity prices synchronization, refer to World Bank 2024 and 2025a.

oil-adjacent commodities have been significantly more responsive to changes in oil prices than under normal market conditions. Implied elasticities of natural gas and fertilizer prices with respect to oil prices are close to 50 percent larger.

Overall, the findings show that oil supply disruptions occurring in turbulent geopolitical contexts have had distinct and particularly destabilizing consequences for oil and other commodity markets. They also indicate that a differentiated assessment of oil market developments is needed during periods of heightened geopolitical instability. The findings also reinforce the case for proactive policy measures to limit the exposure of economies—particularly oil importers—to such shocks.

Methodology and data

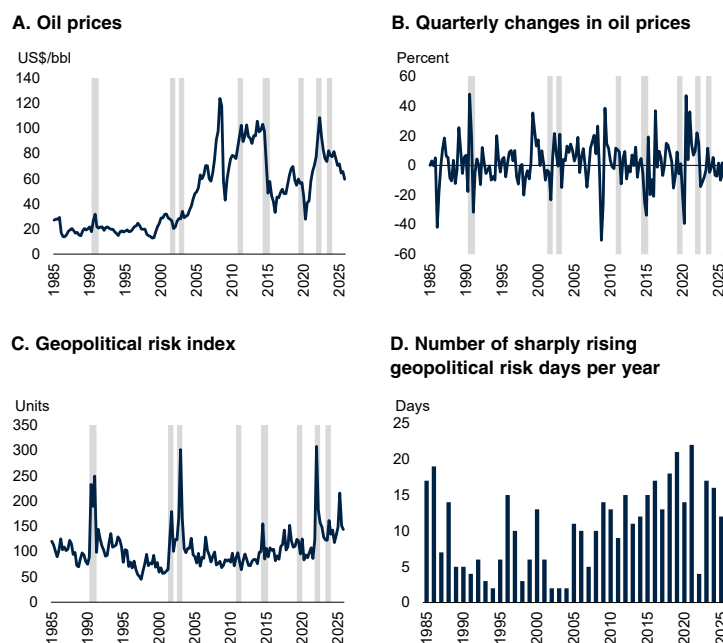
The special focus uses a novel identification strategy to isolate geopolitical shocks to oil markets over the period January 1985–October 2023 and estimate their effects on oil prices and quantities and on other commodity prices.²

Identification. The identification strategy isolates oil price movements that have occurred alongside spikes in geopolitical risk and that are likely to have been exogenous to global macroeconomic conditions. This approach aims to quantify the empirical relationship between spells of rising geopolitical risk and oil prices. Rapid increases in geopolitical tensions have often been associated with sharp and unexpected surges in oil prices, reflecting increased market expectations of future supply disruptions, the materialization of such disruptions, and associated precautionary and speculative demand (figures SF.1.A and SF.1.B).

Geopolitical oil price surprises and oil supply shocks. Geopolitical risk is measured using the threats component of the Geopolitical Risk (GPR) Index developed by Caldara and Iacoviello (2022), which is intended to capture the intensity of geopolitical threats based on reports in leading

FIGURE SF.1 Oil prices and geopolitical risk

Oil prices are especially sensitive to perceived geopolitical risks, often responding to rising geopolitical tensions even in the absence of actual supply disruptions, and rising more sharply still if disruptions materialize. Recent years have seen an increase in the frequency of surges in geopolitical risk, as gauged by references in newspaper articles.



Sources: Caldara and Iacoviello (2022); World Bank.

A–C. Shadow areas highlight nine events associated with significant increases in geopolitical risk and sharp, immediate oil price responses. In chronological order: Iraq's invasion of Kuwait (August 1990); the start of the Gulf War (January 1991); the terrorist attacks in the United States (September 2001); the oil strike in República Bolivariana de Venezuela (December 2002); the start of the civil war in Libya (February 2011); the breakdown of OPEC cooperation amid escalation of conflicts in Ukraine and the Syrian Arab Republic (November 2014); the drone attacks on Saudi Arabia's oil facilities (September 2019); Russia's invasion of Ukraine (February 2022); and the outbreak of conflict in the Middle East following attacks on Israel (October 2023).

C. Geopolitical risk index (threats category) developed by Caldara and Iacoviello (2022). It captures references in international newspapers to geopolitical tensions signaling potential escalation of conflict. These include mentions of diplomatic frictions, threats of military action, war warnings, and related language indicating heightened geopolitical risk. Using a keyword-based methodology applied to 11 major international newspapers, Caldara and Iacoviello (2022) identify relevant articles and compute the monthly share of such references relative to total news coverage.

D. A day of sharply rising geopolitical risk is defined as a day in which the geopolitical risk index increases by more than two standard deviations of its daily average price change (an increase of close to 200 percent). 426 such days were identified out of 14,961 days during 1985–2025, representing 2.8 percent of all days.

international newspapers (figure SF.1.C).³ Days of rising geopolitical risk are identified as those in which the GPR index increases by more than 2 standard deviations of its daily growth rate (an increase of close to 200 percent; figure SF.1.D). For these episodes, changes in oil prices are com-

² Recent comprehensive analyses of the relationship between geopolitical developments and oil price movements include Baumeister and Hamilton (2023), Baumeister (2023), Bondarenko et al. (2024), Kilian, Plante, and Richter (2024), and Pinchetti (2024). A full literature review is provided by Verduzco-Bustos and Zanetti (2026).

³ The *Threats* component of the index is used in the analysis because it reflects a broad range of events that affect expectations or risk premia in energy markets, not just events that affect actual supply conditions, which may have been anticipated and already reflected in market prices. News related to the likelihood of geopolitical events tends to affect oil prices through expectations and risk premia, while the subsequent occurrence of those events may have limited impact if they are already priced in by markets.

puted within a narrow event window encompassing the trading days before and after the event. A data series of daily geopolitical oil price surprises is thereby constructed, and a monthly series of such surprises is obtained by aggregating the daily surprises over each month. This measure is subsequently used as an external instrument in a proxy vector autoregression (VAR) framework to identify the effects of geopolitical shocks on oil market variables, thereby isolating geopolitical oil supply shocks.⁴

Spillover analysis. Following the identification of geopolitical oil supply shocks using the proxy SVAR framework, local projection methods are applied to examine spillovers to other commodity markets. This approach provides a flexible and transparent framework for estimating impulse responses, and is particularly well suited to capturing heterogeneous and country-specific transmission patterns of geopolitical oil supply shocks.

Data and sources. Monthly data are used for the sample period January 1985 to October 2023. The benchmark model includes three core oil market variables. Global oil production is measured in thousands of barrels per day and obtained from the U.S. Energy Information Administration (EIA). The real oil price is measured using the West Texas Intermediate (WTI) crude oil price deflated by the U.S. Consumer Price Index (CPI), obtained from the Federal Reserve Economic Data (FRED). Oil inventories are petroleum inventories as reported by the Organisation for Economic Co-operation and Development (OECD), following Kilian and Murphy (2014). Additional control variables are also included in the model to account for broader market and macroeconomic conditions.⁵

For the analysis of spillover effects, monthly data are employed for seven commodity price indexes—the aggregate World Bank commodity price index and six subindices covering fertilizers,

foods, metals and minerals, natural gas, precious metals, and agricultural raw materials—and for six individual commodities. In all, price data for 29 commodities are included in the analysis. This disaggregation allows a systematic assessment of heterogeneous responses across commodity categories with different production structures and degrees of energy intensity. All commodity price data series are obtained from the World Bank *Pink Sheet*, which provides consistent and internationally comparable monthly price data.

Behavior of oil prices in periods of rising geopolitical risk

As a starting point for assessing oil price behavior during periods of rising geopolitical risk, the volatility of oil prices is compared between months characterized by sharply increasing geopolitical risk perceptions and months when geopolitical risk is viewed as stable (figure SF.2.A). Over the past four decades (1985–2025), oil price volatility (measured by the standard deviation of the monthly growth rates of oil prices) during months of rising geopolitical risk is found to have been about 1.8 times that during stable-risk months. This is consistent with the view that oil price movements tend to be pronounced when geopolitical events occur and may also be amplified by a background environment of generally heightened perceptions of geopolitical risk. Similar patterns are found when the sample is divided into 10-year subperiods. The contrast in volatilities is particularly pronounced in 2006–15, when oil price volatility during mounting risk periods was approximately 2.4 times that during stable-risk periods. During 2016–25, oil price volatility during stable-risk periods was higher than in earlier decades, likely reflecting disruptions associated with the COVID-19 pandemic. However, volatility during months of rising geopolitical risk was again notably greater.

Individual historical events also indicate clearly how geopolitical developments can stoke oil price volatility (figure SF.2.B). Evidence from nine major geopolitical episodes is considered: Iraq's

⁴ Additional details on the construction of the instrument are provided in the annex of this special focus. For further methodological details, refer to Verduzco-Bustos and Zanetti (2026).

⁵ Further details on data construction and sources are provided in Verduzco-Bustos and Zanetti (2026).

invasion of Kuwait (August 1990); the start of the Gulf War (January 1991); the terrorist attacks in the United States (September 2001); the oil strike in República Bolivariana de Venezuela (December 2002); the start of the civil war in Libya (February 2011); the breakdown of OPEC cooperation during the escalation of conflicts in Ukraine and the Syrian Arab Republic (November 2014); the drone attacks on Saudi Arabia's oil facilities (September 2019); Russia's invasion of Ukraine (February 2022); and the outbreak of conflict in the Middle East following attacks on Israel (October 2023). Across these episodes, oil prices exhibited an average absolute change of 12 percent within a narrow window of 1 to a few days around each event, with the largest change—33 percent—in January of 1991 following the start of the Gulf War.

Taken together, this evidence suggests that oil price volatility is systematically higher in turbulent geopolitical contexts, with geopolitical events generating oil price movements that are large, abrupt, and unpredictable. This feature is central to the empirical strategy developed in the next section, which seeks to identify the dynamic effects of geopolitical oil supply shocks on oil and other commodity markets.

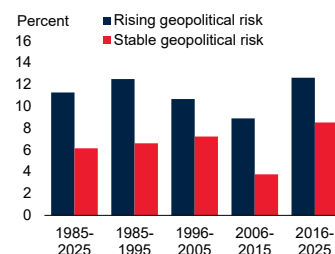
Effects of geopolitical oil supply shocks on the oil market

The dynamic effects of geopolitical oil supply shocks are examined using impulse response functions (IRFs) based on a shock normalized to raise the real price of oil by 10 percent on impact, estimated from the proxy SVAR model (figure SF.3). This normalization facilitates interpretation and allows the magnitude of responses to be compared across variables and over time. The normalized geopolitical oil supply shock is characterized by a contraction in global oil production, leading to a persistent and statistically significant increase in the real price of oil (figures SF.3.A and SF.3.B). This fits standard, widely documented patterns of negative oil supply shocks.

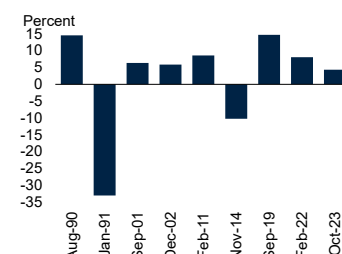
FIGURE SF.2 Oil price volatility around periods of rising geopolitical risk

Between 1985 and 2025, oil price volatility was nearly twice as high during periods of rising geopolitical risk as during low-risk periods, a pattern that remains robust across ten-year subperiods and was particularly pronounced during 2006–15. All nine major geopolitical events identified in the period were associated with large and abrupt oil price movements, with an average absolute price change of about 12 percent. Overall, the evidence indicates that significant increases in geopolitical risk have been systematically associated with unusually large oil price movements.

A. Oil price volatility



B. Oil price changes in the days surrounding major geopolitical risk events



Sources: Caldara and Iacoviello (2022); World Bank.

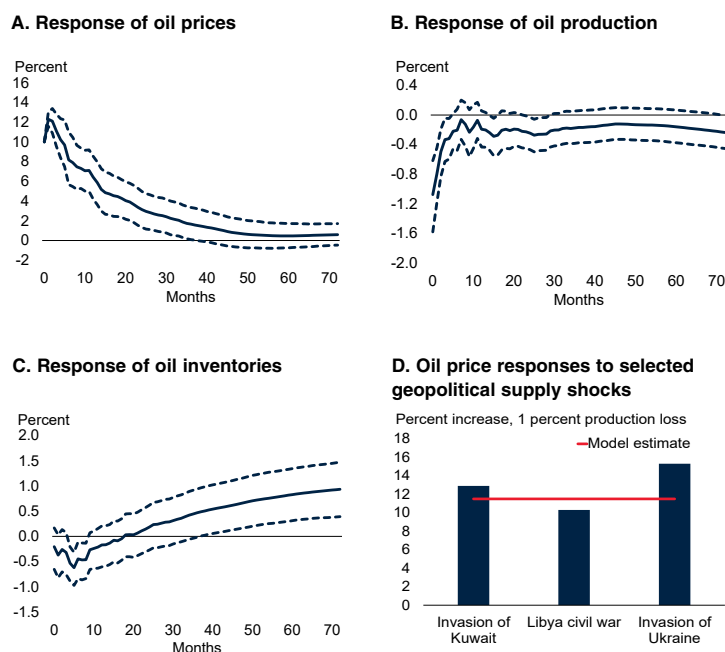
A. Figure shows the average standard deviation of the monthly growth rate of oil prices for different samples. Rising geopolitical risk periods are identified when the monthly growth rate of the geopolitical risk index is two standard deviations from its mean. Stable geopolitical risk periods are defined when the growth rate of the geopolitical risk index is within a band of plus/minus 1 percent.

B. Nine major episodes of high geopolitical risk are identified: Iraq's invasion of Kuwait (August 1990); the start of the Gulf War (January 1991); the terrorist attacks in the United States (September 2001); the oil strike in República Bolivariana de Venezuela (December 2002); the start of the civil war in Libya (February 2011); the breakdown of OPEC cooperation amid escalation of conflicts in Ukraine and the Syrian Arab Republic (November 2014); the drone attacks on Saudi Arabia's oil facilities (September 2019); Russia's invasion of Ukraine (February 2022); and the outbreak of conflict in the Middle East following attacks on Israel (October 2023). Values reflect absolute oil price changes in a narrow window of one or a small number of days surrounding the event, depending on the nature and timing of the event.

The estimated behavior of oil inventories, however, differs from standard supply-shock dynamics. In the short run, inventories decline as they are drawn down to mitigate the supply shortfall, in line with conventional analysis (refer to, for example, Baumeister and Hamilton 2019; Baumeister and Peersman 2013; Kilian 2014; Kilian and Murphy 2014). However, beyond the short term, unlike in most previous studies, inventories not only regain their initial levels within two years but subsequently continue rising. This likely reflects increased precautionary demand for inventories as a form of self-insurance, given heightened perceptions of uncertainty about future supply prospects generated by geopolitical shocks (figure SF.3.C). This is consistent with the mechanism proposed by Känzig (2021), whereby adverse news about future oil supply induces stockpiling.

FIGURE SF.3 Estimated oil market responses to geopolitical oil supply shocks

Around swift rises in geopolitical risk, sharp oil price increases are typically accompanied by declines in production. Inventories are initially drawn down to smooth the short-run production shortfall but rebuild over the medium term and reach higher-than-initial levels. This likely reflects increased precautionary demand amid heightened uncertainty caused by the geopolitical shock.



Sources: International Energy Agency; World Bank.

A.-C. Impulse responses of variables to a shock normalized to raise the real price of oil by 10 percent on impact, with one-standard-deviation confidence bands (dotted lines) computed from 5,000 bootstrap replications.

D. Bars show the first-month oil price increase per 1 percent loss of global oil production during selected historical events; line shows the maximum price response to a 1 percent oil supply shock based on the proxy SVAR model.

Another difference from the results of previous studies is that the estimated response of oil prices to changes in oil production is markedly larger. A 1 percent decline in oil production associated with a geopolitical oil supply shock is estimated to result, on average, in a peak increase of 11.5 percent in oil prices. This oil price impact is nearly twice as large as the estimates in earlier studies using conventional supply-shock identification strategies (Baumeister and Hamilton 2019; Caldara et al. 2019).⁶ The difference reflects the fact that standard approaches typically utilize several sources of supply variation, including natural disasters and

OPEC policy actions. In contrast, by isolating oil price movements linked to geopolitical risk surges, the present analysis allows a more tailored assessment of the specific effects of supply shocks linked to geopolitical events. In terms of behavioral and market drivers, particularly large initial oil price responses to geopolitical supply shocks may be traced to factors that push up risk premia, including urgent efforts to secure physical supplies (or near-term deliveries) immediately after a shock, as well as speculative market responses.

The modeled price response to geopolitical shocks is also consistent with observed historical price responses to material oil supply disruptions linked to geopolitical stress (World Bank 2023). For example, during Iraq's invasion of Kuwait in August 1990, real oil prices increased from \$14.28 to \$25.42 per barrel, while global oil production declined by 6 percent—suggesting a price rise of 12.9 percent for a 1-percent reduction in production. During the civil war in Libya in February of 2011, real oil prices rose from \$40.4 to \$46.1 per barrel, while production fell by 1.38 percent, implying a 10.3-percent increase for a 1 percent supply loss. Similarly, during Russia's invasion of Ukraine in February 2022, real oil prices increased from \$43.9 to \$51.5 per barrel, while global production declined by 1.1 percent, suggesting a price hike of 15.3 percent for a 1 percent production decline. The alignment of the relationship between price and production movements during these historical episodes with the model-based results supports that the modeled geopolitical supply shock effectively captures the dynamics of oil markets during major geopolitical disruptions (figure SF.3.D).

Spillovers to other commodity markets

Oil price movements propagate to other commodity markets through several channels. First, complementarities in extraction and geography mean that negative oil supply shocks tend to coincide with reductions in the supply of related energy commodities, particularly natural gas. Oil and gas are frequently jointly produced because their reserves are co-located, so supply disruptions

⁶ Previous studies find that a 1 percent decrease in oil production is associated with an increase in oil prices of between 4 and 7 percent.

in one market naturally spill over to the other. Second, although the prevalence of oil-indexed pricing has declined, some natural gas contracts maintain explicit or implicit links to oil prices, creating a direct pricing channel. Third, substitution possibilities in energy use can reinforce co-movement: higher oil prices may increase the relative attractiveness of biofuels for transport or of natural gas and coal for heating, raising demand and prices for these commodities and, in the case of biofuels, for underlying agricultural inputs. Finally, energy prices (oil, natural gas, and coal, in different dimensions) exert broad cost-push pressures across the commodity complex. As a key input into transportation, processing, and the production of intermediate goods such as fertilizers and petrochemicals, higher energy costs raise marginal production costs in agriculture, metals, and other sectors, transmitting oil shocks broadly. Together, these channels generate both direct and indirect spillovers from oil prices to other commodity prices.

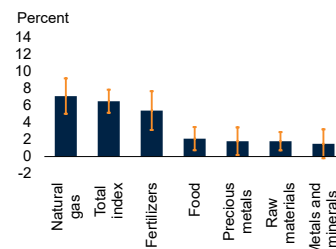
To examine empirically the links between geopolitical oil supply shocks to other commodity prices, impulse responses are estimated for seven commodity price indexes—the aggregate World Bank Group commodity price index, and subindexes for fertilizers, food, metals and minerals, natural gas, precious metals, and agricultural raw materials. Impulse responses are additionally estimated for six individual commodities—European natural gas, urea, grains, aluminum, silver, and timber—selected because of their relative importance in those indexes or as inputs into downstream production.

Geopolitical oil supply shocks have been associated with significant increases in several commodity prices. Oil-intensive commodities, such as natural gas and fertilizers, exhibit the largest price increases, reflecting their strong production complementarities and cost linkages with crude oil. At their peak, normalized to an increase in oil prices of 10 percent on impact, natural gas prices increase, on average, by about 7 percent, and fertilizer prices by slightly more than 5 percent (figures SF.4.A and SF.4.B). Within the natural gas index, European natural gas drives the price surges, with

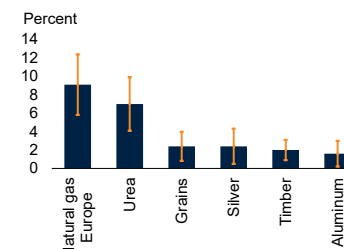
FIGURE SF.4 Spillovers from geopolitical oil supply shocks to other commodity markets

Oil-intensive commodities, such as natural gas and fertilizers, display the largest price responses to geopolitical oil supply shocks. This reflects strong input and cost linkages with crude oil markets and rapid pass-through along supply chains.

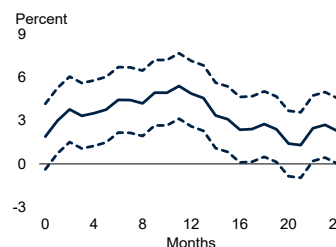
A. Peak responses of non-oil commodity price indexes



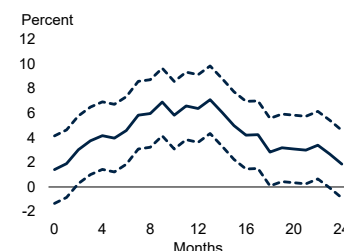
B. Peak responses of selected commodity prices



C. Time path of response of natural gas prices



D. Time path of response of fertilizer prices



Source: World Bank.

A.B. Bars show the largest temporal price deviation across commodity markets following a 10 percent increase in oil prices produced by a geopolitical oil supply shock. Red whiskers indicate one-standard-deviation confidence bands. The 29 commodities included in the indexes are: aluminum, banana, beef, chicken, cocoa, copper, gold, groundnut oil, lamb, logs, natural gas (U.S. and Europe), nickel, orange, phosphate rock, platinum, potassium chloride, rubber, sawnwood, silver, soybean meal, sugar, tea, Thai rice, tin, tobacco, urea, wheat, and zinc.

C.D. Impulse responses of natural gas and fertilizer prices to a geopolitical oil supply shock normalized to raise the real price of oil by 10 percent on impact, with one-standard-deviation confidence bands (dotted lines).

a peak response of more than 9 percent. This underscores Europe's relatively high reliance on gas imports and concomitant exposure to geopolitical energy market shocks. Other commodity groups and individual commodities—including the subindexes for food, agricultural raw materials, and precious metals—also show significant price responses to geopolitical oil supply shocks. Because energy is a key input into the production and transportation of many of these commodities, movements in oil prices are transmitted along the supply chain. However, these price responses have generally peaked at only around 2 percent, implying economically substantial knock-on impacts only in the case of large geopolitical oil shocks.

The estimated time paths of the responses of natural gas and fertilizer prices are also illuminating, given large peak price impacts and the economic significance of these commodities. Price responses for both commodity types build over the year following the shock, peaking at 11 months after the shock for natural gas and at 12 months for fertilizers (figures SF.4.C and SF.4.D). Subsequent price declines similarly exhibit persistence, with prices remaining significantly elevated after 18 months and remaining above their initial levels for more than two years. The gradual nature of the responses following an oil price spike indicates the presence of transmission lags through production, contract structures, and input cost channels. It also underscores that the economic reverberations of large geopolitical oil supply shocks can last for some time, even if much of the initial jump in oil prices reverses relatively swiftly.

These findings are consistent with well-established evidence of substantial co-movement between oil prices and prices of oil-intensive or adjacent commodities. However, the estimates presented here also suggest that such commodity prices become significantly more responsive to oil prices during periods of rising geopolitical risk. On average, based on estimates from previous studies, the implied elasticities of natural gas and fertilizer prices with respect to oil prices are close to 50 percent larger following geopolitical oil supply shocks compared with during normal market conditions (Baffes 2007; Hasanli 2024; World Bank 2016, 2019, 2024).⁷ This apparent amplification indicates that geopolitical oil supply shocks not only alter oil market dynamics but also intensify spillovers to other commodity markets, reinforcing cross-commodity linkages during periods of heightened uncertainty (Baffes 2007; Hasanli 2024; World Bank 2016, 2019, 2024).

⁷ These studies estimate how commodity prices respond to reduced-form fluctuations in oil prices. With respect to fertilizers, Baffes (2007) finds a price response of 0.33 percent to a 1 percent increase in oil prices. For natural gas, Hasanli (2024) estimates an average price response of 0.44 percent (the author reports substantial variation across subsamples, with an estimated response of 0.76 percent for the 1997–2008 period and 0.12 percent for 2009–22).

Conclusions and policy implications

Oil supply shocks associated with geopolitical tensions differ systematically from other types of supply shocks, tending to be particularly disruptive to commodity markets. Declines in oil production associated with geopolitical supply shocks give rise to proportionally larger increases in oil prices than other supply shocks, and also generate sizable spillovers to the prices of other key commodities. After initial declines in inventories, geopolitical oil supply shocks have, on average, led to the accumulation of inventories in the medium term, to levels higher than those preceding the shock. It is likely that this medium-term inventory buildup, not seen with other supply shocks, is motivated by precautionary demand generated by increased uncertainty and risk aversion arising episodes of geopolitical turmoil.

Given that geopolitical shocks to oil prices tend to be comparatively severe, their macroeconomic consequences for growth, inflation, and poverty are also likely to be especially adverse. Their initial inflationary impacts are likely to be greater than those of other types of oil shocks, with a greater tendency to lead to tighter financial conditions, both by potentially inducing counter-inflationary monetary policies and by dampening investor sentiment generally. Challenging macroeconomic repercussions are likely to be particularly pronounced among oil-importing countries, where pressure on the terms of trade and sharp increases in import prices could lead to financial or fiscal destabilization. However, even oil exporters could face headwinds to growth given consumer price pressures, higher non-oil production costs, and weaker investor risk appetite, as well as disruptions to oil exports directly related to the shock.

The frequency of surges in geopolitical risk in the 2020s, coupled with the commodity market and macroeconomic implications of geopolitical shocks, has increased the need for policy action on several fronts. In particular, the case for energy-importing economies to proactively pursue policies to reduce exposure to commodity market disruptions has become stronger. Key priorities include

diversifying energy sources and suppliers, promoting energy efficiency, and developing domestic alternatives to energy imports. The declining costs of renewable energy technologies and storage solutions reinforce the case for expanding renewable electricity generation, helping to reduce geopolitical vulnerability as well as carbon emissions. When energy shocks occur, well-designed fuel pricing frameworks and targeted support for vulnerable households can help limit the pass-through of oil price spikes to domestic inflation and activity.

Oil exporters also face policy challenges associated with the increased frequency of geopolitical shocks. Volatility in export prices can complicate macroeconomic management and fiscal policy, underscoring the importance of countercyclical fiscal rules and prudent debt management to help smooth revenue volatility and support macroeconomic stability. Over the longer term, diversification of production, export destinations, suppliers of key inputs, and transport routes can reduce exposure to price shocks, geopolitical bottlenecks, and trade disruptions. In both oil exporters and importers, deep and transparent energy markets—with robust price information and effective hedging instruments—can help firms and governments manage price volatility (Baffes, Nagle, and Streifel 2026; World Bank 2025b).

Beyond oil markets, the implications of geopolitical supply shocks may increasingly affect a wide range of commodities. Markets for heavily traded commodities—such as liquefied natural gas, fertilizers, and critical minerals—are variously exposed to geopolitical tensions and supply concentration. In some cases, including many industrial metals, global inventories are thin compared with oil, amplifying the potential price and supply chain consequences of disruptions. As the shift away from coal in energy markets, the electrification of transport, and digitalization raise demand for these inputs, susceptibility to geopolitical shocks may evolve without diminishing. This underscores the need for policy efforts to identify and mitigate specific supply vulnerabilities while fostering deep and diversified markets for commodities of mounting importance.

Annex

High-frequency instrument. Following Känzig (2021), Pinchetti (2024), and related high-frequency identification approaches, the analysis uses daily data on oil futures prices to capture the immediate response of oil markets to surges in geopolitical risk. High-frequency data allow for a clearer separation between oil price movements driven by geopolitical developments and those driven by slower-moving macroeconomic factors.

A series of geopolitical oil price surprises is constructed by measuring oil price changes around days when the Geopolitical Risk Index increases sharply. Specifically, for the daily growth rate of the index, a threshold of two standard deviations from the mean (an increase in the index of close to 200 percent) is used to identify days of rising geopolitical tensions and compute the change in oil futures prices over a narrow event window surrounding these episodes (figure SF.1.B). The geopolitical oil price surprise series is defined as the log difference between the price of the one-month-ahead oil futures contract on the business day following the event and the business day preceding it.

One-month-ahead oil futures prices are used because they incorporate market expectations about future supply and demand conditions, storage costs, interest rates, and risk premia. The one-month maturity contract is selected because it provides a reliable measure of short-term oil price dynamics and has the longest available time series among short-horizon contracts. Given the high correlation of oil futures prices across maturities, the results are not sensitive to the specific contract choice.

Aggregation to monthly frequency. For the econometric analysis, the daily geopolitical oil price surprises are aggregated to the monthly frequency following the approach in Känzig (2021). The monthly measure is constructed as follows:

- When a single geopolitical risk event occurs within a month, the monthly surprise equals the corresponding daily surprise.

- When multiple events occur within the same month, the monthly surprise is defined as the sum of the daily surprises.
- When no event occurs, the monthly surprise is set to zero.

This aggregation preserves the timing and magnitude of geopolitical oil price shocks while allowing the analysis to be conducted at the monthly frequency commonly used in macroeconomic applications.

Econometric framework

Proxy SVAR model. The series of geopolitical oil price surprises is used as an external instrument in a structural vector autoregression (SVAR) framework to identify the effects and cross-market spillovers of a structural geopolitical oil supply shock. The analysis in the special focus is centered on the dynamic responses of a set of commodity market variables. The model also includes a vector of additional macroeconomic and financial control variables: world industrial production, measured using an extended version of the OECD's monthly industrial production index that covers OECD countries and six additional major economies, following Baumeister and Hamilton (2019); macroeconomic uncertainty, proxied by the index developed by Jurado, Ludvigson, and Ng (2015); financial uncertainty, proxied by corporate bond credit spreads, as constructed by Gilchrist and Zakrajšek (2012); and, finally, U.S. industrial production, the U.S. Consumer Price Index, and the yield on 10-year constant maturity U.S. Treasury securities (used as a measure of U.S. interest rates),

all obtained from the Federal Reserve Economic Data database.

The identification strategy follows the external instrument (proxy SVAR) approach developed by Stock and Watson (2012) and Mertens and Ravn (2013). This approach exploits the fact that the geopolitical oil price shock series provides a relevant proxy for structural oil price shocks driven by geopolitical developments, while being orthogonal by construction to other structural shocks in the system.

This framework allows for the identification of the dynamic effects of geopolitical oil supply shocks without imposing restrictive contemporaneous zero or sign restrictions. It is therefore well suited for analyzing the transmission of oil price shocks originating from geopolitical developments, while rigorously controlling for macroeconomic variables.

Local projection methods. Once the geopolitical oil price shock is identified using the proxy SVAR framework, local projection methods are employed to extend the analysis to other commodity markets. Conditioning on an orthogonalized shock allows local projections to trace the dynamic responses of a wide range of variables without imposing additional cross-equation restrictions. This approach offers a flexible and transparent way to estimate impulse responses, is robust to potential misspecification of the underlying data-generating process, and is particularly well suited for capturing heterogeneous transmission patterns of geopolitical oil price shocks. The local projection model includes global industrial production and oil prices as control variables.⁸

⁸ For further methodological details refer to Verduzco-Bustos and Zanetti (2026).

References

Executive summary and commodity market developments and outlook

- Arroyo-Marioli, F., J. Khadan, F. Ohnsorge, and T. Yamazaki. 2023. "Forecasting Industrial Commodity Prices: Literature Review and a Model Suite." Policy Research Working Paper 10611, World Bank, Washington, DC.
- Baffes, J., P. Nagle, and S. Streifel. 2025. "International Commodity Agreements and Cartels: Lessons and Policy Implications." *World Bank Research Observer* 40 (2): 290–324.
- Caldara, D., and M. Iacoviello. 2021. "Measuring Geopolitical Risk." International Finance Discussion Paper 1222r1, Board of Governors of the Federal Reserve System, Washington, DC.
- Caldara, D., M. Iacoviello, P. Molligo, A. Prestipino, and A. Raffo. 2019. *Does Trade Policy Uncertainty Affect Global Economic Activity?* Washington, DC: World Bank.
- Consensus Economics. 2026. *Energy & Metals Consensus Forecasts*. March. London: Consensus Economics Inc.
- EIA (U.S. Energy Information Administration). 2025. *About One-Fifth of Global Liquefied Natural Gas Trade Flows through the Strait of Hormuz*. Washington, DC: World Bank.
- EIA (U.S. Energy Information Administration). 2026. *Short-Term Energy Outlook*. Washington, DC: World Bank.
- Energy Institute. 2025. *Statistical Review of World Energy Database*. London: Energy Institute. <https://www.energyinst.org/statistical-review>.
- FAO (Food and Agriculture Organization of the United Nations). 2023. FAOSTAT Database. Rome: FAO. Accessed on April 20, 2026. <https://www.fao.org/faostat/en/#data/RFB>
- Ha, J., M. A. Kose, and F. Ohnsorge. 2023. *One-Stop Source: A Global Database of Inflation*. Journal of International Money and Finance 137: 102896.
- IEA (International Energy Agency). 2014. *Energy Supply Security: Emergency Response of IEA Countries*. Paris: IEA.
- IEA (International Energy Agency). 2024. *Coal 2024*. Paris: IEA.
- IEA (International Energy Agency). 2025. *Coal 2025*. Paris: IEA.
- IEA (International Energy Agency). 2025. *Global EV Outlook 2025*. Paris: IEA.
- IEA (International Energy Agency). 2025. *World Energy Investment 2025*. Paris: International Energy Agency.
- IEA (International Energy Agency). 2026. *Gas Market Report, Q1 2026*. Paris: IEA.
- IEA (International Energy Agency). 2026. *Oil Market Report*. Paris: IEA.
- IEA (International Energy Agency). 2026. *Strait of Hormuz Factsheet*. Paris: IEA.
- IGC (International Grains Council). 2026. *Grain Market Report*. GMR 574. London: IGC.
- International Fertilizer Association (database). *Fertilizer Statistics Database*. Paris: International Fertilizer Association. Accessed on April 15, 2026. <https://www.ifastat.org/>.
- Loungani, P., E. Luttini, and H. Pallan. 2025. "Buffering Recessions: Labor Market Asymmetries and the Role of Self-Employment." Policy Research Working Paper 11089, World Bank, Washington, DC.
- North Atlantic Treaty Organization. 2025. *Defence Expenditure of NATO Countries (2014–2025)*. Brussels: NATO.
- Refinitiv (database). "Refinitiv Commodities and Financial Markets Database." Accessed on March 25, 2026. <https://www.lseg.com/en/data-analytics/trading-solutions/world-bureau-metal-statistics>.
- U.S. Department of Agriculture (database). *Production, Supply, and Distribution Online (PSD Online)*. Accessed on April 15, 2026. <https://apps.fas.usda.gov/psdonline/>.

- U.S. Department of Agriculture, National Agricultural Statistics Service. 2026. *Prospective Plantings*. Washington, DC: USDA NASS.
- WFP (World Food Programme). 2025. *WFP 2026 Global Outlook: Hunger and Hope: Innovative Solutions to Address Food Insecurity*. Rome: WFP.
- WFP (World Food Programme). 2026. *Projected Increase in Acute Food Insecurity Due to the Middle East Conflict*. Rome: Rome: WFP.
- World Bank. 2020. *Minerals for Climate Action: The Mineral Intensity of the Clean Energy Transition*. Washington, DC: World Bank.
- World Bank. 2023. *Commodity Markets Outlook: Under the Shadow of Geopolitical Risks*. Washington, DC: World Bank.
- World Bank. 2026. *Global Economic Prospects*. January. Washington, DC: World Bank.
- Special focus**
- Baffes, J. 2007. "Oil Spills on Other Commodities." *Resources Policy* 32 (3): 126–34.
- Baffes, J., P. Nagle, and S. S. Streifel. 2026. "Policy Lessons from International Commodity Agreements: Failure of Non-Oil Pacts and the Endurance of OPEC." Policy Research Working Paper 11340, World Bank, Washington, DC.
- Baumeister, C. 2023. "Pandemic, War, Inflation: Oil Markets at a Crossroads?" NBER Working Paper 31496, National Bureau of Economic Research, Cambridge, MA.
- Baumeister, C., and J. D. Hamilton. 2019. "Structural Interpretation of Vector Autoregressions with Incomplete Identification: Revisiting the Role of Oil Supply and Demand Shocks." *American Economic Review* 109 (5): 1873–910.
- Baumeister, C., and J. D. Hamilton. 2023. "Uncovering Disaggregated Oil Market Dynamics: A Full-Information Approach to Granular Instrumental Variables." Mimeo.
- Baumeister, C., and G. Peersman. 2013. "Time-Varying Effects of Oil Supply Shocks on the U.S. Economy." *American Economic Journal: Macroeconomics* 5 (4): 1–28.
- Bondarenko, Y., V. Lewis, M. Rottner, and Y. Schüler. 2024. "Geopolitical Risk Perceptions." *Journal of International Economics* 152 (November): 104005.
- Caldara, D., M. Cavallo, and M. Iacoviello. 2019. "Oil Price Elasticities and Oil Price Fluctuations." *Journal of Monetary Economics* 103 (May): 1–20.
- Caldara, D., and M. Iacoviello. 2022. "Measuring Geopolitical Risk." *American Economic Review* 112 (4): 1194–225.
- Gilchrist, S., and E. Zakrajšek. 2012. "Credit Spreads and Business Cycle Fluctuations." *American Economic Review* 102 (4): 1692–720.
- Hasanli, M. 2024. "Re-Examining Crude Oil and Natural Gas Price Relationship: Evidence from Time-Varying Regime-Switching Models." *Energy Economics* 133 (May): 107510.
- IEA (International Energy Agency). 2026. *Oil Market Report*. March. Paris: International Energy Agency.
- Jurado, K., S. C. Ludvigson, and S. Ng. 2015. "Measuring Uncertainty." *American Economic Review*, 105 (3): 1177–216.
- Känzig, D. R. 2021. "The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements." *American Economic Review* 111 (4): 1092–125.
- Kilian, L. 2014. "Oil Price Shocks: Causes and Consequences." *Annual Review of Resource Economics* 6 (1): 133–54.
- Kilian, L., and D. P. Murphy. 2014. "The Role of Inventories and Speculative Trading." *Journal of Applied Econometrics* 29 (3): 454–78.
- Kilian, L., M. Plante, and A. W. Richter. 2024. "Geopolitical Oil Price Risk and Economic Fluctuations." Working paper.
- Mertens, K., and M. O. Ravn. 2013. "The Dynamic Effects of Tax Changes." *American Economic Review* 103 (4): 1212–47.
- Pinchetti, M. 2024. "Deconstructing Geopolitical Risk Surprises." Working paper.
- Stock, J. H., M. W. Watson, A. S. Blinder, and C. A. Sims. 2012. "Disentangling the Channels of the

2007–09 Recession.” *Brookings Papers on Economic Activity*: 81–156.

Verduzco-Bustos, G., and F. Zanetti. 2026. “The Effects of Geopolitical Oil Price Shocks.” CEPR Discussion Paper No. 21378, CEPR Press, Paris and London.

World Bank. 2016. “Resource Development in an Era of Cheap Commodities.” In *Commodity Markets Outlook*, 11–19. April. Washington, DC: World Bank.

World Bank. 2019. “Food Price Shocks: Channels and Implications.” In *Commodity Markets Outlook*, 5–16. April. Washington, DC: World Bank.

World Bank. 2023. “Potential Near-Term Implications of the Conflict in the Middle East for Commod-

ity Markets: A Preliminary Assessment.” In *Commodity Markets Outlook*, 11–19. October. Washington, DC: World Bank.

World Bank. 2024. “Commodity Price Synchronization: A New Era?” In *Commodity Markets Outlook*, 39–51. October. Washington, DC: World Bank.

World Bank. 2025a. “Post-Pandemic Commodity Cycles: A New Era?” In *Commodity Markets Outlook*, 41–57. April. Washington, DC: World Bank.

World Bank. 2025b. “New Shocks, Old Tools: Revisiting International Commodity Agreements in a Fragmented World.” In *Commodity Markets Outlook*, 37–55. October. Washington, DC: World Bank.